POOJA KIRAN

Security Engineer | AI & Agent Security | Application Security | Cloud IAM

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Professional Experience

Independent AI Security Researcher & Engineer Aug 2024 – Present

Self-Directed Engineering | Tempe, AZ

- Built Python security tooling for MCP tool authorization, IAM policy analysis, and model artifact inspection across independently developed security projects.
- Implemented fail-closed decisions, structured findings, audit trails, and CI-based regression testing with pytest, GitHub Actions, CodeQL, and Docker.

Business & Compliance Lead | AEROSEC Aug – Dec 2025

Technology Innovation Lab Externship | Arizona State University | Externally mentored aviation project

- Defined third-party Passenger Service System and vendor API risks for a proposed airline security dashboard with compliance and incident-reporting workflows.
- Compared four commercial security solutions and developed a proposed $120,000 first-year budget with a 12-month runway model.

Graduate Teaching Assistant | IT Grader Jan – Oct 2025

Ira A. Fulton Schools of Engineering | Arizona State University

- Assessed approximately 85 undergraduate Web Programming submissions per semester and 52 graduate cybersecurity assignments using instructor rubrics.
- Provided technical feedback on secure development, security compliance, risk analysis, and information-systems security coursework.

Security Engineering Projects

MCP Agent Security Gateway | Python | FastAPI | MCP/JSON-RPC | SIEM Oct 2025 – Oct 2026

- Engineered a default-deny agent-to-tool gateway with capability checks, request inspection, anti-SSRF egress rules, and rate limits.
- Implemented 69 named prompt-injection signatures, PII/exfiltration signals, hash-chained auditing, and nine Elastic Security rule records.
- Built automated regression coverage for gateway controls and SIEM integration; the Python 3.12 suite passed 782 tests with 81.93% statement coverage.

AWS Agent Identity Guard | Python | AWS IAM | SARIF 2.1.0 Apr 2025 – Oct 2026

- Built a static workload-identity analyzer for dangerous grants, escalation paths, trust-policy weaknesses, and missing permission boundaries.
- Implemented 25 deterministic rules for wildcard access, iam:PassRole, sts:AssumeRole, and trust policies; emitted CI-readable SARIF findings.
- Tested IAM policy analysis through CI, with 244 passing tests; three live-AWS checks were skipped because they required credentials.

HF Model Provenance Scanner | Python | SafeTensors | GGUF | ONNX | Keras Nov 2024 – Oct 2026

- Built non-executing model artifact scanning for pickle-derived, SafeTensors, GGUF, ONNX, and Keras formats.
- Combined serialization, AST/taint, provenance, and dependency checks; flagged 33 of 33 adversarial fixtures.
- Tested model artifact inspection in CI: 260 tests passed with 75.67% statement coverage.

Attack Detection Engine | Firmware & CAN Security | Python | Embedded C | ARM Aug 2026 – Oct 2026

- Extended offline CAN frame analysis, C/C++ firmware security checks, and ARM ELF32 inspection using controlled test fixtures.
- Cross-compiled Cortex-M3 firmware; reproduced and fixed an intentional C stack overflow using AddressSanitizer.
- Optimized ATT&CK index test setup; full Python 3.12 regression suite passed 68/68 tests in 23.42 seconds.

Technical Skills

Programming & engineering: Python, Bash, Rust, C++; FastAPI, pytest, Docker, GitHub Actions

Security engineering: MCP/JSON-RPC, tool authorization, API security, anti-SSRF, prompt injection, threat modeling, IAM

DevSecOps & analysis: AWS IAM, SARIF 2.1.0, CodeQL, Trivy, Bandit, pip-audit, Elastic Security, SIEM, Prometheus

Education

Arizona State University | M.S., Information Technology (Security) | May 2026 | Tempe, AZ

M. S. Ramaiah University of Applied Sciences | B.Tech., Computer Science & Engineering | Aug 2023 | Bengaluru, India

Certifications

AWS Academy Graduate: Cloud Architecting (Apr 2025) | Cloud Security Foundations (Nov 2025)